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CO2-Assignment Tool 2

SCENARIO BASED

Course code: DSA0308
Course Name: Natural Language Processing
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1. Morphological Processing for Search Engine Indexing

Introduction

Morphological processing is an NLP method for examining how words are formed. It identifies a word's root and affixes, such as prefixes and suffixes.

Inflectional morphology changes grammatical form without normally changing the word class, as in analyze → analyzing. Derivational morphology creates a related form and may change its grammatical role, as in analyze → analytical.

Problem Statement

A search engine may receive queries containing words such as analyzing, analysis, and analytical. If each form is indexed separately, relevant documents may be missed. A rule-based program is therefore required to identify the specified word components, classify the forms, and map them to a shared representation for retrieval.

Aim

To create a Python program that applies predefined morphological rules to analyzing, analysis, and analytical and prints their root, suffix, transformation type, and normalized form.

Python Program

```
words = ["analyzing", "analysis", "analytical"]

rules = {

    "analyzing":  ("analyze", "ing",    "Inflectional"),

    "analysis":    ("analysis", "None", "Base/related form"),

    "analytical": ("analysis", "tical", "Derivational")

}

for word in words:

    root, suffix, category = rules[word]

    normalized = "analysis"

    print("Word:", word)

    print("Root:", root)

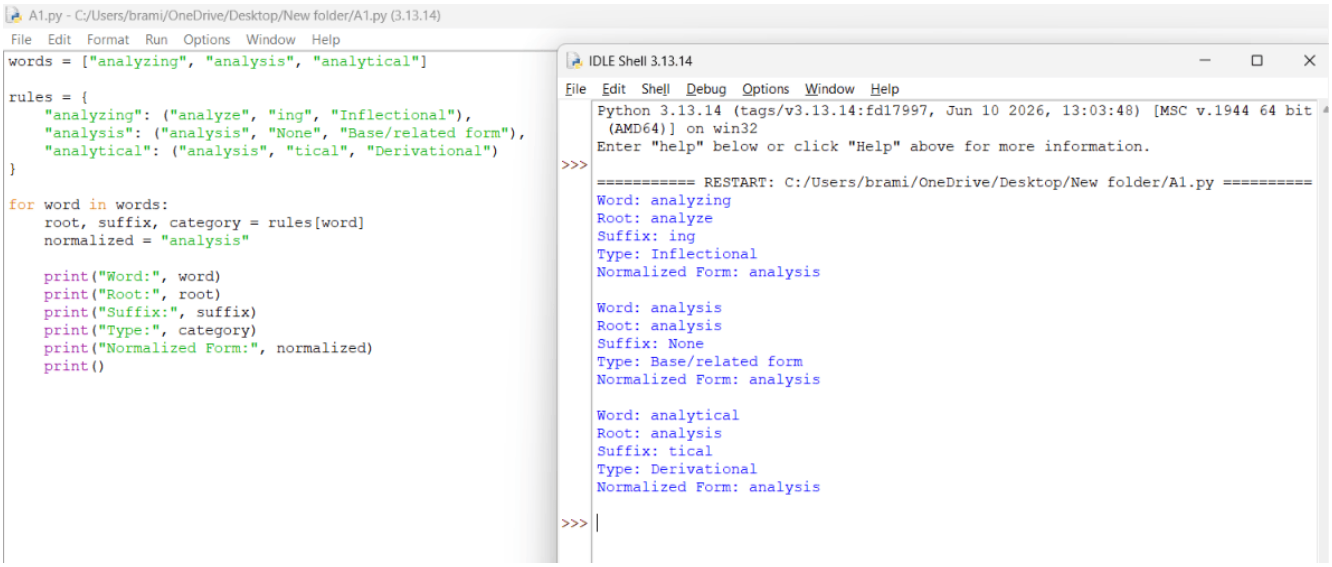
    print("Suffix:", suffix)
```

```
print("Type:", category)

print("Normalized Form:", normalized)

print()
```

Sample Output



Solution Approach

The input words are stored in a list, and a dictionary provides the predefined analysis for each word. A loop retrieves the information and prints a normalized form. The mapping is an illustrative, word-specific rule set rather than a general morphological analyzer. In particular, analysis and analyze are related forms, but they are not produced from one another by simply removing the suffix -ing.

Result

The Python program displays the predefined root, suffix, category, and normalized representation for all three input words. This demonstrates how a rule-based mapping can group related terms for search indexing.

2. Morphology-Based Word Normalization Using Python

Introduction

Morphological normalization examines related words and maps them to a shared base representation. Derivational suffixes can form new words from a root. For example, govern is a verb, while government and governance are related nouns.

Problem Statement

A news analytics pipeline receives the words govern, government, and governance. Although their meanings are related, their endings and word classes differ. A program is needed to identify their roots, suffixes, word classes, and derivational levels, and then normalize them to a common representation.

Aim

To implement a Python program that analyzes govern, government, and governance and maps them to the shared base govern for text organization and retrieval.

Python Program

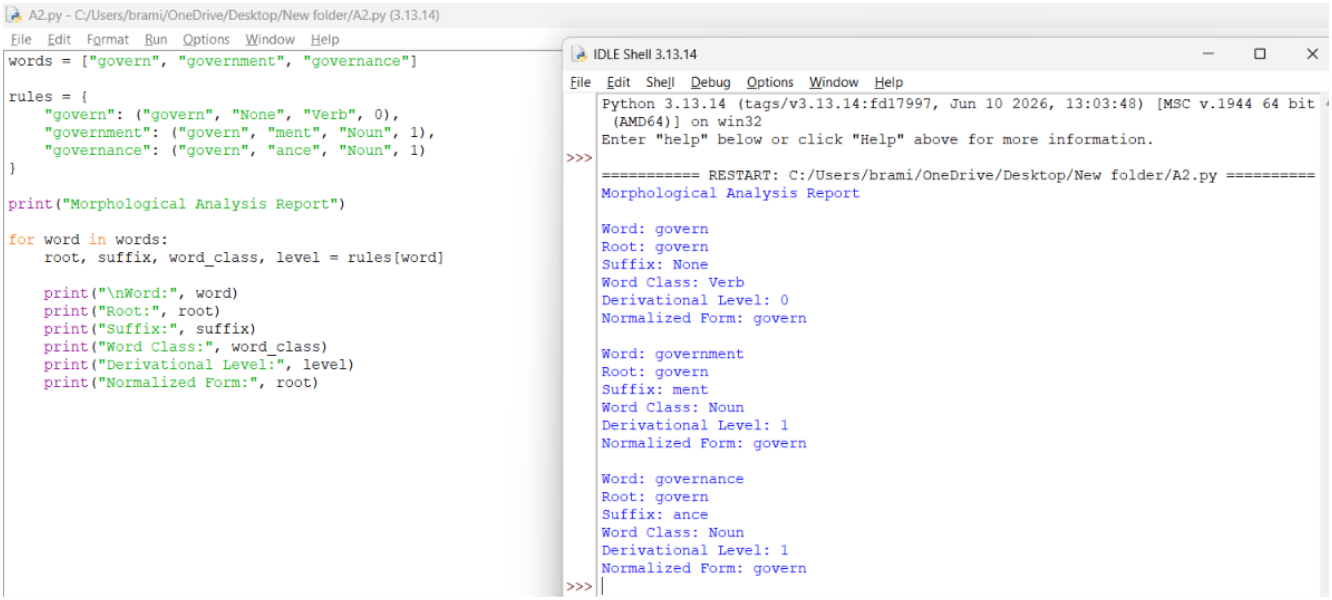
```
words = ["govern", "government", "governance"]

rules = {
    "govern":     ("govern", "None", "Verb", 0),
    "government": ("govern", "ment", "Noun", 1),
    "governance": ("govern", "ance", "Noun", 1)
}

print("Morphological Analysis Report")

for word in words:
    root, suffix, word_class, level = rules[word]
    print("\\nWord:", word)
    print("Root:", root)
    print("Suffix:", suffix)
    print("Word Class:", word_class)
    print("Derivational Level:", level)
    print("Normalized Form:", root)
```

Sample Output



Solution Approach

The words are placed in a list, while a dictionary stores each word's root, suffix, word class, and derivational level. The loop prints these details and uses the root as the normalized form. The suffix labels and levels are predefined for this small example; the code does not infer them automatically for unknown words.

Result

The program identifies the supplied morphological details and maps govern, government, and governance to govern. This shared representation can help group related news articles for categorization and document clustering.